

ATAM-FET: An Adaptive Toxic Attention Framework for Federated Edge-Based Early Toxicity Detection on Mobile Keyboards

Saida Binte Alam, Najmus Saquib Aurko, Anindita Roy

Faculty of Science and Technology

American International University-Bangladesh (AIUB)

Dhaka, Bangladesh

{26-94136-2, 26-94142-2, 26-94077-2}@student.aiub.edu

Abstract—Toxic and abusive posts have multiplied alongside the growth of social media, chat applications, and public forums, and moderating them by hand no longer scales. Automated detectors help, but the usual design choice running them on a central server forces users to upload their text for scanning. That choice raises clear privacy questions, leans on expensive server hardware, and typically reacts only once a message is already public, so nothing is done to stop harm before it happens. We are working toward ATAM-FET, a federated, on-device framework that screens text for toxicity directly on the mobile keyboard. Its centrepiece is an Adaptive Toxic Attention Module (ATAM) that sharpens toxicity-related context before a decision is made, together with federated learning and edge-side model optimization so that no raw text leaves the phone. This paper covers the first step of that agenda: a centralized, multi-label DistilBERT classifier trained on the Jigsaw Toxic Comment corpus (159,571 comments across six labels-toxic, severe toxic, obscene, threat, insult, and identity hate). With an 80/10/10 stratified split and a full cleaning and WordPiece-tokenization routine, the model reaches a subset (exact-match) accuracy of 91.5%, a micro-averaged F1-score of 0.767, a macro-averaged F1-score of 0.679, a ROC-AUC of 0.984, and a Hamming loss of 0.019 on the held-out test set. Breaking the scores down by label shows dependable results on common categories and weaker ones on the rare labels, which exposes the effect of heavy class imbalance and makes the case for ATAM. These findings give us a firm reference point for the ATAM, federated-learning, and edge-optimization stages that follow.

Index Terms—Multi-label toxicity detection, DistilBERT, transformer fine-tuning, class imbalance, content moderation, federated learning, edge computing, privacy-preserving NLP.

I. INTRODUCTION

Online conversation now happens at a scale that would have been hard to imagine a decade ago: more than 5.31 billion people are active on social platforms, and many of them run into hostile content. The reported figures are sobering. Close to 41% of internet users say they have faced online harassment, and roughly one student in three has been bullied on the web. The damage does not stop at a single insult. Repeated exposure to hostile language has been tied to anxiety, depression, withdrawal from social life, and, in the worst cases, self-harm [22], [23]. For these reasons, detecting toxic language, hate speech, and cyberbullying automatically has

grown into an active and socially consequential line of research [4], [13].

Most current detectors are built on pre-trained transformers like BERT and its many relatives because they read context, sarcasm, and shades of meaning that lexicon- or feature-based classifiers routinely miss [3], [24], [26]. Putting such models into a live moderation service, however, runs into three recurring obstacles. The first is privacy: a server-side pipeline has to receive the user’s text. Which are some of the most personal content a person writes before it can act, which sits awkwardly with user expectations and with rules such as the GDPR [8], [10], [20]. The second is cost, since leading transformer models are hungry for memory, computation, and battery, none of which a phone has to spare [7], [10]. The third is timing: because moderation usually fires only after a message is posted, the harmful text has generally already reached its target before anything is flagged.

Our answer to these obstacles is ATAM-FET (Adaptive Toxic Attention Module for Federated Edge-based Toxicity Detection), a longer-term research programme that pairs small transformer models with toxicity-aware attention, federated learning, and edge-side optimization so that screening can run on the keyboard itself. The attention component, ATAM, is meant to push the model’s focus toward the tokens that actually signal toxicity, which we expect to help most on the under-represented categories while keeping the computation light enough for a handset. ATAM is a planned design in this paper and has not yet been built.

What we report here is the opening stage: a solid centralized DistilBERT baseline for multi-label toxicity classification on the Jigsaw corpus. The work covers the whole path from raw text to prediction cleaning, WordPiece tokenization, fine-tuning, and a careful reading of the outcome through multi-label metrics, a per-class breakdown, and the training curves. ATAM, federated learning, and the edge-side steps (quantization, pruning, and knowledge distillation) belong to later stages and are set out as future work. The baseline in this paper is the yardstick those additions will be measured against.

The main contributions of this paper are as follows:

- We put forward ATAM-FET, to our knowledge the first

design aimed at catching toxic text while it is still being typed on a mobile keyboard. Rather than judging a message after it is submitted, the framework is meant to flag it beforehand and keep the text on the device, opening the door to intervention before any harm is done.

- We describe the Adaptive Toxic Attention Module (ATAM), a compact attention mechanism that re-weights contextual token representations by how strongly they relate to toxic language, with the aim of catching subtle and rare categories without inflating the compute budget.
- We lay out an end-to-end, privacy-preserving edge pipeline that ties DistilBERT together with ATAM, Federated Averaging (FedAvg), and edge-side optimization (knowledge distillation, quantization, and pruning), so that a shared model can improve without any raw user text being collected.
- We implement and study the centralized baseline in depth and set out the protocol for the remaining stages, in which the full framework will be tested across centralized, federated, and edge-deployment settings with ablations, per-class analysis, and communication-cost and latency measurements.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the proposed framework and the centralized baseline methodology. Section IV describes the experimental setup. Section V reports and discusses the experimental results. Section VI outlines the planned implementation of the Adaptive Toxic Attention Module, federated learning, and edge optimization. Finally, Section VII concludes the paper.

II. RELATED WORK

A. Transformer-Based Toxicity and Hate-Speech Detection

Work on spotting toxic and hateful text has moved from simple keyword lists and classical classifiers toward deep networks and, more recently, transformers [4]. The survey by Jahan and Oussalah [4] maps this shift and shows how thoroughly contextual language models now dominate. Several studies confirm that BERT-style models beat recurrent and lexical baselines. Chandavarkar et al. [3] pair Bi-LSTM and CNN layers with a compact TinyBERT for toxic-comment labelling, and Almaliki et al. [5] show that a slimmed-down Arabic BERT-Mini works well outside English. Saleh et al. [26] report that combining BERT with hate-speech-tuned embeddings beats strong deep baselines, and ensembles push results further still: Yoo et al. [9] blend several BERT variants through parallel model fusion for Korean–English detection, while Mazari et al. [25] stack BiLSTM and BiGRU heads on BERT for multi-aspect labelling. Hashmi and Yildirim Yayilgan [11] mix FastText features with a BiLSTM–GRU network and multilingual transformers to handle low-resource Norwegian, and Agarwal et al. [12] speed detection up with parallelized ensembles. The literature also reaches well beyond English: Roy et al. [31] build a deep ensemble for Dravidian languages, and Kapil and Ekbal [32] tackle multimodal hate speech with

multi-task learning. Ayetiran and Özgöbek [33] route several modalities through a shared inter-modal attention scheme for joint fake-news, hate, and offensive-language detection, whereas Anand et al. [34] combine feature selection with an ensemble classifier. Yi and Zubiaga [35], surveying session-based cyberbullying, argue that a large part of the signal lives in the conversational context. Further threads cross-lingual transfer [1], [14], [15], [17], multi-modal fusion [6], user-profile cues [28], explainable moderation [27], embedding choices [29], and attempts to define hate speech operationally [30] round out the picture. Across all of them the accuracy of transformers is not in doubt, but every method assumes a central data store and server-grade hardware.

B. Federated Learning for Privacy-Preserving Text Moderation

Federated learning (FL) lets a group of clients train one shared model without pooling their data; only the model updates travel to a coordinator, which merges them with schemes such as FedAvg [8], [10]. Surveys have charted its use in decentralized and edge scenarios [20], [21]. Within content moderation, Abdel Samee et al. [8] bring together word embeddings, emotion features, and a BERT client under FL for cyberbullying detection, reaching about 92% accuracy for a federated BERT while keeping each platform’s data private and adding a differential-privacy analysis. Leonidou et al. [2] present a working privacy-preserving moderation use case, and meta-learning has been tried for few-shot, low-resource abusive text [1], [19]. Closer to our setting, Nagy et al. [36] adapt privacy-preserving FL specifically to language tasks, and Khan et al. [37] review the FL-for-NLP landscape, cataloguing its architectures, datasets, and open problems. Evidence from neighbouring fields, such as the healthcare framework of Wang et al. [38], suggests that sharing updates under formal privacy guarantees carries over to other kinds of sensitive data. The common thread is that FL can approach centralized accuracy without giving up data locality, yet these systems mostly keep full-size models and stop short of on-keyboard, before-send screening.

C. Communication-Efficient and Lightweight Federated Learning

The sticking point when transformers meet FL is the size of the model that has to be shipped every round. Galal et al. [7] sidestep much of it with FedFreezeBERT, freezing BERT and training only a small aggregation head, which trims communication by as much as $6.2\times$ against FedAvg and still raises accuracy. Wu et al. [10] take a different route with FedKD, using two-way knowledge distillation and adaptive gradient compression to cut communication by up to 94.89% while holding centralized-level accuracy on non-IID data. A parallel line of work distills, quantizes, and prunes compact backbones such as DistilBERT, MobileBERT, and TinyBERT [3], [11]. For transformers specifically, Zhang et al. [39] pair dual-feature distillation with progressive pruning to shrink BERT without much accuracy loss, and Hong et al. [40] chart

how distillation trades size for performance. On the systems side, edge-oriented FL keeps improving: Zhang et al. [41] add Byzantine robustness on top of communication savings for mobile-edge networks, Herabadi [42] balances training across uneven IoT-edge clients with a semi-synchronous hierarchy, and Zhou et al. [43] fold in dataset distillation for federated edge learning that is both lean and attack-resistant. Read together, these results make one point plain: small models and aggressive compression are not optional for edge FL, and they shape the roadmap in Section VI. Table I summarizes the methods discussed above by theme, core methodology, and contribution.

D. Research Gap

Read together, the three strands above tell a consistent story. Transformers classify toxic text well; FL keeps training data private; and compression brings model sizes down to something a device can carry. What no one has yet done is bring all three together for detection that runs early, on the keyboard, on the phone itself. This paper takes the first concrete step in that direction by fixing a careful centralized multi-label baseline on which the federated, compressed, on-device system can later be built and against which it can be compared.

III. PROPOSED METHODOLOGY

A. Problem Formulation

We frame toxicity detection as a *multi-label* problem: a single comment x can carry more than one of $L = 6$ labels at once (toxic, severe toxic, obscene, threat, insult, identity hate). The model f_θ turns x into L independent probabilities $\hat{y} \in [0, 1]^L$ through a sigmoid layer, and a label l is switched on when $\hat{y}_l \geq 0.5$. Training minimizes the mean binary cross-entropy across the labels:

$$\mathcal{L} = -\frac{1}{L} \sum_{l=1}^L [y_l \log \hat{y}_l + (1 - y_l) \log(1 - \hat{y}_l)]. \quad (1)$$

B. Target Deployment Architecture

Figure 1 presents the complete end-to-end workflow of the proposed ATAM-FET framework. The framework encompasses the entire development pipeline, beginning with data preprocessing and centralized model training, followed by evaluation, and extending toward the planned deployment stages including the Adaptive Toxic Attention Module (ATAM), federated learning, and edge-oriented model optimization through quantization, pruning, and knowledge distillation.

The internal architecture of the proposed neural network is illustrated in Figure 2. The model is built upon the DistilBERT encoder and is customized by introducing the proposed Adaptive Toxic Attention Module (ATAM) between the transformer encoder and the multi-label classification head. ATAM learns toxicity-aware attention weights over contextual token embeddings to produce a toxicity-aware contextual representation before classification, enabling the model to emphasize

semantically important toxic expressions while maintaining a lightweight architecture suitable for mobile devices.

Figure 3 illustrates the target federated deployment architecture. In the complete ATAM-FET framework, the customized model will execute locally on mobile keyboards, performing early toxicity detection while users are typing. Each client will train using its own private data and transmit only encrypted model updates to a central server, where Federated Averaging (FedAvg) will aggregate the updates into a global model without collecting raw user text.

The scope of this paper is limited to the first stage of the proposed framework. Specifically, we implement and evaluate the centralized DistilBERT baseline that serves as the foundation for ATAM-FET. The Adaptive Toxic Attention Module, federated learning pipeline, and edge optimization techniques remain planned extensions and are discussed in Section VI.

C. DistilBERT Backbone

For the baseline encoder we use **DistilBERT**, a distilled BERT of roughly 66 million parameters and six transformer layers that keeps about 97% of BERT’s language ability at a fraction of the cost. That trade-off is exactly what a future on-device deployment needs.

On top of the encoder we place a small head: one fully connected layer feeding six independent sigmoid units, one per toxicity label. This is the reference model for ATAM-FET. Later work will slot the Adaptive Toxic Attention Module (ATAM) in between the encoder and this head so that toxicity-related context is emphasized before the final decision.

We fine-tune the encoder and head together on the Jigsaw training split, running six epochs of AdamW with a learning rate that decays linearly from 2×10^{-5} to 0, the multi-label cross-entropy of (1), and a 128-token limit. At test time a fixed threshold of 0.5 turns probabilities into labels.

Everything reported below refers to this centralized baseline alone. ATAM, federated learning, quantization, pruning, and knowledge distillation arrive in later stages of ATAM-FET and are therefore outside the current evaluation.

IV. EXPERIMENTAL SETUP

A. Dataset and Statistics

Our data is the Jigsaw Toxic Comment Classification set 159,571 comments labelled by people against six binary toxicity tags. It is badly skewed: only about one comment in ten carries any tag at all, and *threat*, the scarcest, shows up in just 0.30%. Table II lists the split sizes and Table III the per-label counts. We drop 282 exact duplicates and divide the remaining 159,289 comments 80/10/10 with a stratified split on the toxic/non-toxic distinction. Jigsaw’s separate 153,164-comment test collection is set aside for the later federated phase and plays no part here.

B. Preprocessing

Cleaning proceeds in a fixed order: strip URLs and @-mentions with regular expressions, turn emojis into text through `emoji.demojize()`, lowercase everything, and

TABLE I
COMPARISON OF HATE SPEECH DETECTION METHODOLOGIES

Category	Reference	Year	Core Methodology	Key Contribution
Transformer-Based Detection	Chandavarkar et al. [3]	2024	Hybrid Bi-LSTM, CNN, TinyBERT	Combined recurrent and convolutional layers with compact transformers
	Almaliki et al. [5]	2023	Arabic BERT-Mini	Demonstrated efficiency of compact BERT for non-English detection
	Yoo et al. [9]	2025	Parallel Model Fusion	Leveraged ensemble techniques for multilingual (Korean–English) robustness
	Mazari et al. [25]	2024	BERT + BiLSTM / BiGRU	Explored multi-aspect detection using ensemble-based backbones
	Agarwal et al. [12]	2023	Parallelized Ensembles	Focused on accelerating detection speeds via parallelized learning
Federated Learning	Abdel Samee et al. [8]	2023	Federated BERT + Emotional Features	Achieved 92% accuracy while preserving platform privacy
	Leonidou et al. [2]	2023	Federated Learning Use Case	Established practical frameworks for privacy-preserving moderation
Efficiency & Edge Opt.	Galal et al. [7]	2024	FedFreezeBERT	Froze BERT parameters to reduce communication costs by 6.2×
	Wu et al. [10]	2022	FedKD (Knowledge Distillation)	Reduced communication costs by 94.89% via adaptive distillation
This Work	–	2026	DistilBERT + INT8 + FL	Proposed centralized baseline for pre-send toxicity detection

TABLE II
STRATIFIED 80/10/10 DATASET SPLIT.

Split	Comments	Proportion
Training	127,431	80.0%
Validation	15,929	10.0%
Test	15,929	10.0%
Total (after de-duplication)	159,289	100%

TABLE III
PER-LABEL POSITIVE-SAMPLE DISTRIBUTION (OF 159,571 COMMENTS).
AVERAGE LABELS PER COMMENT: 0.22; 16,174 COMMENTS CARRY AT
LEAST ONE LABEL.

Label	Positive comments	Share
toxic	15,294	9.58%
severe_toxic	1,595	1.00%
obscene	8,449	5.29%
threat	478	0.30%
insult	7,877	4.94%
identity_hate	1,405	0.88%

squeeze repeated whitespace; we then drop the 282 duplicates and swap any empty string for a `[no_text]` marker. The cleaned text goes through DistilBERT’s WordPiece AutoTokenizer, is padded or cut to 128 tokens, and is paired with a binary attention mask. About 6.9% of sequences ran past the length cap and were truncated. The six labels become a multi-hot float vector, and the whole thing is wrapped as a Hugging Face DatasetDict.

TABLE IV
FINE-TUNING CONFIGURATION FOR THE CENTRALIZED BASELINE.

Setting	Value
Backbone	DistilBERT (distilbert-base-uncased)
Task head	Linear + 6 sigmoid units (multi-label)
Max sequence length	128 tokens
Optimizer	AdamW
LR schedule	Linear decay, $2 \times 10^{-5} \rightarrow 0$
Epochs	6
Loss	Multi-label binary cross-entropy
Decision threshold	0.5
Training regime	Centralized (single node)

C. Training Configuration

D. Evaluation Metrics

Since the task is both multi-label and skewed, we lean on several metrics rather than one. From the per-label counts of true and false positives and negatives:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad (2)$$

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (3)$$

We give both the micro average, which pools every label–instance decision and therefore tracks the common labels, and the macro average, a plain mean over labels that lays bare how the rare ones fare. We add subset (exact-match) accuracy and

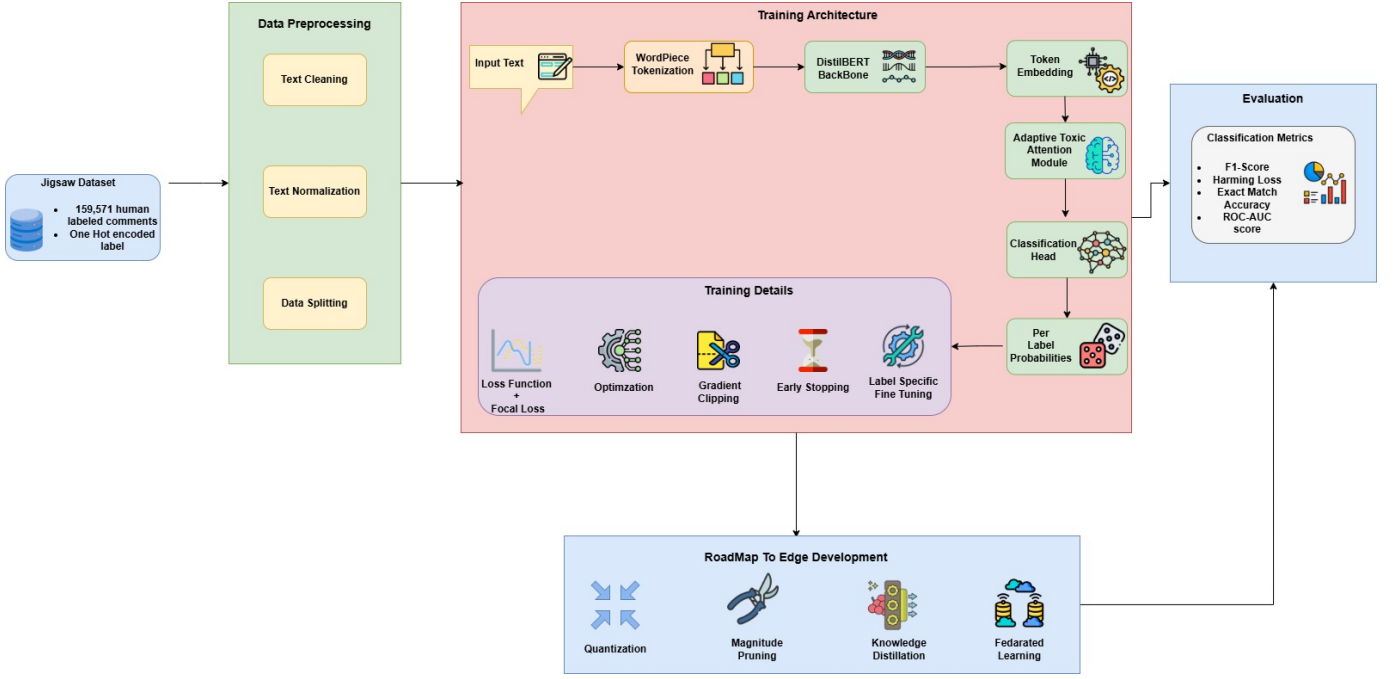


Fig. 1. Overall workflow of the proposed ATAM-FET framework. The pipeline begins with the Jigsaw Toxic Comment dataset, followed by data preprocessing, centralized model training and evaluation. The framework is then extended through the proposed Adaptive Toxic Attention Module (ATAM), federated learning, and edge-oriented optimization techniques including quantization, pruning, and knowledge distillation for deployment on mobile keyboards. The present work implements only the centralized DistilBERT baseline; the remaining stages constitute future work.

Hamming loss over the N test comments and L labels,

$$\text{SubsetAcc} = \frac{1}{N} \sum_i \mathbf{1}[\hat{y}_i = y_i], \quad (4)$$

$$\text{HammingLoss} = \frac{1}{NL} \sum_i \sum_l \mathbf{1}[\hat{y}_{il} \neq y_{il}], \quad (5)$$

and finally the macro-averaged area under the ROC curve (ROC-AUC), which does not depend on the 0.5 threshold.

E. Implementation

The model runs in PyTorch on top of the Hugging Face Transformers library and was trained on a single GPU. That single-GPU limit is why the federated experiments are on hold for now. The code lives in a public repository¹ and is still under active development.

V. RESULTS AND DISCUSSION

A. Test-Set Performance

Table V gives the held-out results over 15,929 comments. The model gets all six labels right on more than nine of every ten comments (91.5% subset accuracy) and misses fewer than 2% of individual label decisions (Hamming loss 0.019). A macro ROC-AUC of 0.984, which ignores the threshold entirely, says the ranking is strong. The distance between the micro-F1 (0.767) and macro-F1 (0.679) is telling: the model does markedly better on frequent labels than on rare ones. Micro-recall (0.864) also sits above micro-precision (0.690),

TABLE V
CENTRALIZED DISTILBERT BASELINE ON THE JIGSAW HELD-OUT TEST SET (15,929 COMMENTS).

Metric	Value	Type
Subset (exact-match) accuracy	0.915	—
Hamming loss	0.019	lower is better
ROC-AUC	0.984	macro
F1-score	0.767	micro
Precision	0.690	micro
Recall	0.864	micro
F1-score	0.679	macro
Precision	0.601	macro
Recall	0.787	macro

so the classifier errs on the side of calling something toxic. It catches most real cases and pays for it with some false alarms, which is the safer failure mode for a protective tool.

B. Training Dynamics and Overfitting

Table VI and Fig. 4 follow the six epochs. On the training side everything improves—loss drops from 0.465 to 0.073 and F1-macro climbs from 0.479 to 0.861—but validation tells another story: its loss rises without pause from 0.299 after epoch 1 to 0.885 by epoch 6, while validation F1-macro flattens out, topping at 0.677 at epoch 5 and slipping to 0.675 at epoch 6. A training curve that keeps improving while the validation curve turns the other way is a plain sign of overfitting. Stopping around the fifth epoch would give the best validation model, which is exactly what the regularization plans in Section VI target.

¹<https://github.com/0NJS0/Toxic-Comment-detection>

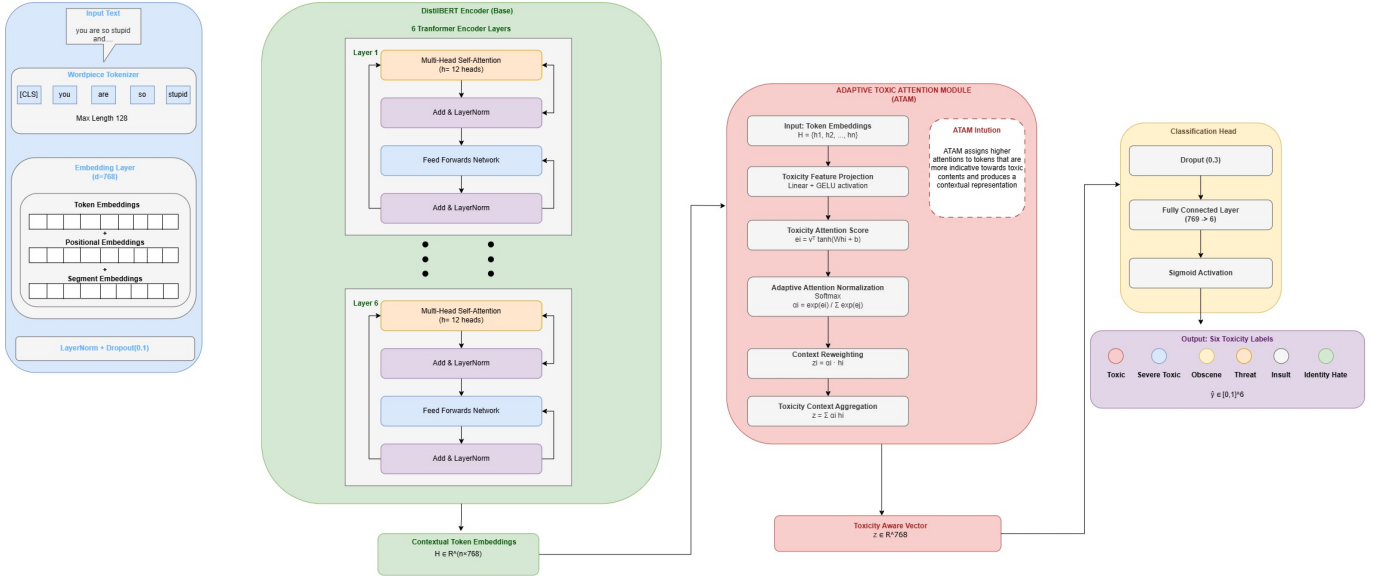


Fig. 2. Architecture of the proposed ATAM-FET model. The model consists of a DistilBERT encoder with six transformer layers, followed by the proposed Adaptive Toxic Attention Module (ATAM), which generates a toxicity-aware contextual representation from contextual token embeddings. This representation is passed to a multi-label classification head that predicts six independent toxicity labels. In the current implementation, only the DistilBERT baseline is trained; ATAM represents the proposed architectural enhancement to be implemented in future work.

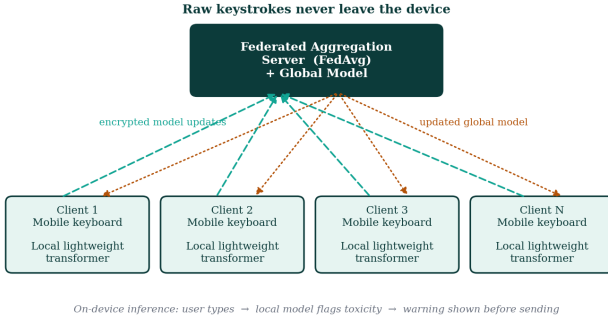


Fig. 3. Target federated, edge-based deployment architecture. Each mobile keyboard runs a local lightweight transformer and shares only encrypted model updates, which a server aggregates with FedAvg. In this paper only the on-device local model (the centralized baseline of Section III-D) is implemented; the federated aggregation loop shown here is future work (Section VI).

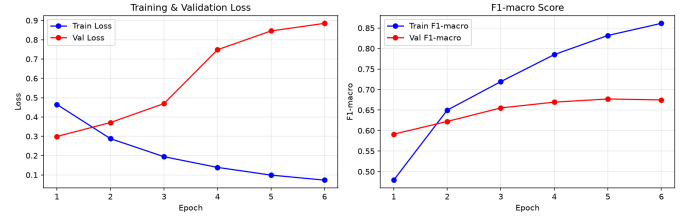


Fig. 4. Training and validation loss (left) and F1-macro (right) over six epochs. Training loss falls while validation loss rises after the first epoch, and validation F1-macro plateaus—clear evidence of overfitting.

0.801, and insult at 0.745. While the three scarce ones lag: severe toxic 0.529, threat 0.587, and identity hate 0.595. The ordering follows label frequency almost exactly, which points straight at class imbalance as the main culprit; with only 478 threat examples (0.30%) to learn from, the model never sees enough to draw a sharp boundary. Recall holds up even on the rare labels (threat recall 0.653), but precision sags (0.533 for threat, 0.413 for severe toxic), so the classifier raises many false positives where examples are thin. That is why the macro-F1 of 0.679 sits well below the micro-F1 of 0.767, which the plentiful labels prop up.

D. Discussion and Limitations

The baseline shows that a small transformer can handle toxicity well on the whole, but three shortcomings set the agenda. It is centralized, so the privacy and before-send goals are still unmet. It struggles on the rare, high-stakes labels (threat, identity hate), and the single 0.5 threshold is not tuned per label. And it overfits past the fifth epoch, so it needs firmer regularization or an earlier stop. No compression has been

TABLE VI

PER-EPOCH TRAINING DYNAMICS. VALIDATION F1-MACRO PEAKS AT EPOCH 5.

Ep.	Tr. loss	Val loss	Tr. acc	Val acc	Val F1 _m
1	0.465	0.299	0.854	0.869	0.591
2	0.288	0.371	0.901	0.897	0.622
3	0.194	0.469	0.921	0.907	0.655
4	0.139	0.748	0.940	0.914	0.669
5	0.099	0.846	0.953	0.916	0.677
6	0.073	0.885	0.961	0.918	0.675

C. Per-Class Analysis and Class Imbalance

Fig. 5 and Table VII split the results by label. The three common tags come out well. Toxic at 0.816 F1, obscene at

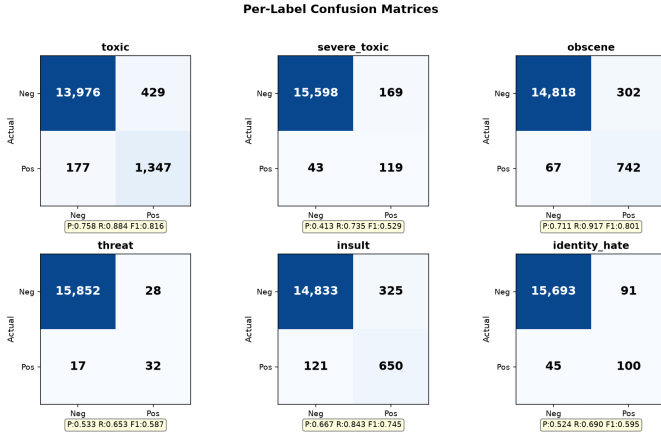


Fig. 5. Per-label confusion matrices on the test set, with per-label precision (P), recall (R), and F1. Frequent labels (toxic, obscene, insult) are detected reliably, whereas rare labels (severe toxic, threat, identity hate) show lower precision.

TABLE VII
PER-LABEL TEST PERFORMANCE (SUPPORT = POSITIVE TEST COMMENTS, FROM THE CONFUSION MATRICES OF FIG. 5).

Label	Precision	Recall	F1	Support
toxic	0.758	0.884	0.816	1,524
severe_toxic	0.413	0.735	0.529	162
obscene	0.711	0.917	0.801	809
threat	0.533	0.653	0.587	49
insult	0.667	0.843	0.745	771
identity_hate	0.524	0.690	0.595	145

applied either, which leaves the footprint too large for a phone. Section VI turns each of these into a concrete next step.

E. Comparison with Prior Work

Table VIII places our baseline next to a set of recent toxicity and hate-speech studies, listing each one’s dataset, model, parameter count, and reported accuracy and F1. The datasets and label schemes differ from paper to paper, so this is a rough orientation rather than a like-for-like contest; our own row is left blank for now and will be filled once ATAM, federated learning, and the edge-optimized models are in place.

VI. PLANNED FRAMEWORK EXTENSIONS

The stages below build directly on the baseline.

A. Adaptive Toxic Attention Module (ATAM)

The centralized DistilBERT model is the springboard for ATAM-FET. Its first extension is the Adaptive Toxic Attention Module (ATAM), meant to sharpen toxicity-related context just before the label decision.

Self-attention inside the transformer already links tokens across a sentence, yet the final decision usually rests on a single pooled vector, which can wash out faint toxic cues, especially for scarce labels like *threat*, *identity hate*, and *severe toxic*. ATAM instead learns its own weights over the token

embeddings so the model can lean on the tokens that matter most for a toxic reading. Let

$$H = \{h_1, h_2, \dots, h_n\}, \quad h_i \in \mathbb{R}^d, \quad (6)$$

denote the contextual token representations from the DistilBERT encoder, where n is the sequence length and d the hidden size. For each token, ATAM produces a toxicity-aware score

$$e_i = \mathbf{v}^T \tanh(W h_i + b), \quad (7)$$

with learnable W , b , and $\mathbf{v}$. A softmax turns the scores into weights,

$$\alpha_i = \frac{\exp(e_i)}{\sum_{j=1}^n \exp(e_j)}, \quad (8)$$

that sum to one,

$$\sum_{i=1}^n \alpha_i = 1, \quad (9)$$

and the toxicity-aware representation is their weighted combination,

$$z = \sum_{i=1}^n \alpha_i h_i. \quad (10)$$

This vector then feeds the classification head,

$$\hat{\mathbf{y}} = \sigma(W_c z + b_c), \quad (11)$$

with $\sigma(\cdot)$ the sigmoid used for multi-label output.

By design, ATAM should draw the model toward the words that carry toxic meaning and away from filler, which we expect to help most on the minority labels. Because it is a planned addition, its build-out and testing are left for future work; we will judge it by comparing DistilBERT with and without ATAM under the same multi-label metrics in an ablation.

B. Federated Training

With ATAM in place, the next stage moves the model into a federated setting. The training data will be spread over simulated mobile clients and combined with Federated Averaging [8], [10]: in round t each chosen client k takes local steps,

$$\theta_k^{t+1} \leftarrow \theta^t - \eta \nabla F_k(\theta^t), \quad (12)$$

and the server forms the next global model as a sample-weighted mean,

$$\theta^{t+1} = \sum_k \frac{n_k}{n} \theta_k^{t+1}, \quad (13)$$

so that the text itself never leaves the phone. We will look at non-IID client splits and at communication-savers such as parameter freezing [7] and gradient distillation or compression [10].

TABLE VIII
QUANTITATIVE COMPARISON WITH REPRESENTATIVE PRIOR WORK ON TOXICITY AND HATE-SPEECH DETECTION

Reference	Year	Dataset	Model / Backbone	Params	Accuracy (%)	F1 (%)
Abdel Samee et al. [8]	2023	Cyberbullying (Kaggle, multi-platform)	Federated BERT + emotional features	~110M	92.15	88.65
Saleh et al. [26]	2023	Davidson; Waseem	BERT + hate-speech embeddings	~110M	–	96.0
Lu et al. [16]	2023	Davidson	Dual contrastive learning (BERT)	~110M	95.0	95.0
Hashmi and Yayilgan [11]	2024	Norwegian hate speech	FAST-RNN (FastText + BiLSTM-GRU)	NR	98.0	98.0
Mazari et al. [25]	2024	Jigsaw / Wikipedia Toxic Comments	BERT + BiLSTM/BiGRU ensemble	~110M	–	99.0
Yoo et al. [9]	2025	Measuring Hate Speech (English)	Ensemble of BERT variants (PMF)	NR	85.0	85.0
This Work	2026	Jigsaw Toxic Comment	DistilBERT (proposed ATAM-FET)	66M	–	–

Note: The values are those reported by the original papers, each measured on its own dataset and label scheme, so the comparison is indicative rather than a controlled head-to-head test. Parameter counts refer to the main transformer backbone where applicable; NR = not reported. The row for *This Work* is left blank and will be completed once the ATAM enhancement, federated learning, and edge-optimized models are trained.

C. Model Compression for Edge Deployment

Once ATAM and federation are working, the model has to fit a phone, so we will apply three compression steps. Knowledge distillation will hand knowledge from the trained DistilBERT teacher down to a smaller student such as TinyBERT or MobileBERT,

$$\mathcal{L}_{KD} = (1 - \alpha) \mathcal{L}_{BCE} + \alpha \tau^2 \text{KL}(p_s^\tau \| p_t^\tau); \quad (14)$$

INT8 post-training quantization will remap 32-bit weights onto 8-bit integers,

$$q = \text{round}(r/s) + z; \quad (15)$$

and magnitude pruning will cut the least useful weights.

Evaluation will proceed one piece at a time. ATAM comes first, measured against the centralized baseline in this paper; the ATAM-enhanced model then passes through federated training and edge optimization, scored on exact-match accuracy, macro- and micro-averaged F1-score, ROC-AUC, model size, communication cost, and on-device inference latency.

D. Planned Ablation Study

Because ATAM, federated learning, and the edge optimizations are not built yet, a standard all-at-once ablation is not possible at this point; instead the framework will be assessed as each part lands, always against the same baseline.

Stage one checks ATAM by putting DistilBERT+ATAM head to head with the plain DistilBERT baseline, with special attention to the scarce labels *threat*, *identity hate*, and *severe toxic*. Stage two adds federated learning through FedAvg to see how decentralized, non-IID training behaves while keeping data private. Stage three brings in quantization, pruning, and distillation to bring down size, memory, and latency for real phones.

Every stage uses the same dataset and protocol as the baseline and is scored on the same battery of metrics exact-match

TABLE IX
PLANNED EVALUATION STAGES OF THE PROPOSED ATAM-FET FRAMEWORK. ONLY THE CENTRALIZED BASELINE IS IMPLEMENTED IN THE CURRENT STUDY.

Stage	Status
Centralized DistilBERT Baseline	Completed
DistilBERT + ATAM	Planned
DistilBERT + ATAM + Federated Learning	Planned
ATAM + FL + Edge Optimization	Planned
Complete ATAM-FET Framework	Planned

accuracy, precision, recall, macro- and micro-averaged F1-score, ROC-AUC, Hamming loss, model size, communication cost, and latency. So that each component’s contribution can be read off cleanly. Table IX lists the stages and their current status.

E. Robustness Improvements

Two problems stand out in the results: overfitting after the fifth epoch and weak numbers on the minority labels *threat*, *identity hate*, and *severe toxic*. We plan to tackle both from two directions. ATAM should help on the architecture side by foregrounding toxic context, so that faint cues in the rare classes are less likely to be lost. On the training side we will try early stopping, class-weighted or focal losses, resampling of the minority classes, and per-label threshold tuning. Each will be tested on its own and in combination against the baseline under the same protocol and metrics.

VII. CONCLUSION

This paper delivered the first stage of ATAM-FET, a federated, edge-based framework aimed at catching toxic text early on mobile keyboards. As the groundwork, we trained a centralized DistilBERT model for multi-label toxicity classification on the Jigsaw Toxic Comment corpus, with the full

pipeline of cleaning, tokenization, and fine-tuning behind it. On the held-out set it reached a subset accuracy of 91.5%, a micro-averaged F1-score of 0.767, a macro-averaged F1-score of 0.679, a ROC-AUC of 0.984, and a Hamming loss of 0.019. A label-by-label look pinned the main weaknesses on class imbalance and overfitting, most visibly for the rare *threat*, *identity hate*, and *severe toxic* categories.

Those numbers point to the next move: adding the Adaptive Toxic Attention Module to lift toxicity-related context before the decision, then extending the model with privacy-preserving federated learning and edge-side optimization, quantization, pruning, and knowledge distillation for on-device use that never ships raw text.

Taken as a whole, the baseline here is the measuring stick for everything that follows, and it charts a workable path toward accurate, private, and lightweight toxicity detection for the next wave of mobile messaging.

REFERENCES

- [1] M. Mozafari, R. Farahbakhsh, and N. Crespi, "Cross-lingual few-shot hate speech and offensive language detection using meta learning," *IEEE Access*, vol. 10, pp. 14880–14896, 2022.
- [2] P. Leonidou, N. Kourtellis, N. Salamanos, and M. Sirivianos, "Privacy-preserving online content moderation: A federated learning use case," in *Companion Proc. ACM Web Conf. (WWW '23)*, 2023, pp. 280–289.
- [3] C. A. Chandavarkar, M. Rayhan, R. Soni, and S. Muthukumar, "Toxic comment classification using Bi-LSTM, CNN, and TinyBERT," in *Lecture Notes in Networks and Systems*. Singapore: Springer Nature, 2024, pp. 525–535.
- [4] M. S. Jahan and M. Oussalah, "A systematic review of hate speech automatic detection using natural language processing," *Neurocomputing*, vol. 546, art. 126232, 2023.
- [5] M. Almaliki, A. M. Almars, I. Gad, and E.-S. Atlam, "ABMM: Arabic BERT-Mini model for hate-speech detection on social media," *Electronics*, vol. 12, no. 4, art. 1048, 2023.
- [6] R. Prabhu and V. Seethalakshmi, "A comprehensive framework for multi-modal hate speech detection in social media using deep learning," *Scientific Reports*, vol. 15, art. 13020, 2025.
- [7] O. Galal, A. H. Abdel-Gawad, and M. Farouk, "Federated Freeze BERT for text classification," *Journal of Big Data*, vol. 11, art. 28, 2024.
- [8] N. Abdel Samee, U. Khan, S. Khan, M. M. Jamjoom, M. Sharif, and D. H. Kim, "Safeguarding online spaces: A powerful fusion of federated learning, word embeddings, and emotional features for cyberbullying detection," *IEEE Access*, vol. 11, pp. 124524–124541, 2023.
- [9] S. Yoo, E. Jeon, J. Hyeon, and J. Cho, "Adaptive ensemble techniques leveraging BERT-based models for multilingual hate speech detection in Korean and English," *Scientific Reports*, vol. 15, art. 19844, 2025.
- [10] C. Wu, F. Wu, L. Lyu, Y. Huang, and X. Xie, "Communication-efficient federated learning via knowledge distillation," *Nature Communications*, vol. 13, art. 2032, 2022.
- [11] E. Hashmi and S. Yildirim Yayilgan, "Multi-class hate speech detection in the Norwegian language using FAST-RNN and multilingual fine-tuned transformers," *Complex & Intelligent Systems*, vol. 10, pp. 4535–4556, 2024.
- [12] S. Agarwal, A. Sonawane, and C. Ravindranath Chowdary, "Accelerating automatic hate speech detection using parallelized ensemble learning models," *Expert Systems with Applications*, vol. 230, art. 120564, 2023.
- [13] S. Raza, "HarmonyNet: Navigating hate speech detection," *Natural Language Processing Journal*, vol. 8, art. 100098, 2024.
- [14] A. Firmino, B. Bezerra, and A. C. A. Nascimento, "Improving hate speech detection using cross-lingual learning," *Expert Systems with Applications*, vol. 235, art. 121115, 2024.
- [15] L. Liu, K. Xu, and B. Liu, "A cross-lingual transfer learning method for online COVID-19-related hate speech detection," *Expert Systems with Applications*, vol. 234, art. 121031, 2023.
- [16] J. Lu, B. Xu, X. Zhang, C. Min, L. Yang, and H. Lin, "Hate speech detection via dual contrastive learning," *IEEE/ACM Trans. Audio, Speech, and Language Processing*, vol. 31, pp. 2787–2795, 2023.
- [17] A. Khan et al., "Offensive language detection for low-resource language using deep sequence model," *IEEE Trans. Computational Social Systems*, 2023.
- [18] K. Nithya et al., "Hybrid approach of deep feature extraction using BERT-OPCNN & FIAC with customized Bi-LSTM for rumor text classification," *Alexandria Engineering Journal*, vol. 90, pp. 65–75, 2024.
- [19] M. R. Awal, R. K.-W. Lee, E. Tanwar, T. Garg, and T. Chakraborty, "Model-agnostic meta-learning for multilingual hate speech detection," *IEEE Trans. Computational Social Systems*, vol. 11, no. 1, pp. 1086–1095, 2024.
- [20] Q. W. Khan, A. N. Khan, A. Rizwan, R. Ahmad, S. Khan, and D.-H. Kim, "Decentralized machine learning training: A survey on synchronization, consolidation, and topologies," *IEEE Access*, vol. 11, pp. 68031–68050, 2023.
- [21] S. Khan, A. Rizwan, A. N. Khan, M. Ali, R. Ahmed, and D. H. Kim, "A multi-perspective revisit to the optimization methods of neural architecture search and hyper-parameter optimization for non-federated and federated learning environments," *Computers & Electrical Engineering*, vol. 110, art. 108867, 2023.
- [22] M. Vismara, N. Girone, D. Conti, G. Nicolini, and B. Dell'Osso, "The current status of cyberbullying research: A short review of the literature," *Current Opinion in Behavioral Sciences*, vol. 46, art. 101152, 2022.
- [23] G. W. Giumetti and R. M. Kowalski, "Cyberbullying via social media and well-being," *Current Opinion in Psychology*, vol. 45, art. 101314, 2022.
- [24] R. Ali, U. Farooq, U. Arshad, W. Shahzad, and M. O. Beg, "Hate speech detection on Twitter using transfer learning," *Computer Speech & Language*, vol. 74, art. 101365, 2022.
- [25] A. C. Mazari, N. Boudoukhani, and A. Djeflal, "BERT-based ensemble learning for multi-aspect hate speech detection," *Cluster Computing*, vol. 27, pp. 325–339, 2024.
- [26] H. Saleh, A. Alhothali, and K. Moria, "Detection of hate speech using BERT and hate speech word embedding with deep model," *Applied Artificial Intelligence*, vol. 37, no. 1, art. 2166719, 2023.
- [27] H. Mehta and K. Passi, "Social media hate speech detection using explainable artificial intelligence (XAI)," *Algorithms*, vol. 15, no. 8, art. 291, 2022.
- [28] G. del Valle-Cano, L. Quijano-Sánchez, F. Liberatore, and J. Gómez, "SocialHaterBERT: A dichotomous approach for automatically detecting hate speech on Twitter through textual analysis and user profiles," *Expert Systems with Applications*, vol. 216, art. 119446, 2023.
- [29] M. Umer et al., "Impact of convolutional neural network and FastText embedding on text classification," *Multimedia Tools and Applications*, vol. 82, no. 4, pp. 5569–5585, 2023.
- [30] J. Papcunová, M. Martončík, D. Fedáková, et al., "Hate speech operationalization: A preliminary examination of hate speech indicators and their structure," *Complex & Intelligent Systems*, vol. 9, no. 3, pp. 2827–2842, 2023.
- [31] P. K. Roy, S. Bhawal, and C. N. Subalalitha, "Hate speech and offensive language detection in Dravidian languages using deep ensemble framework," *Computer Speech & Language*, vol. 75, art. 101386, 2022.
- [32] P. Kapil and A. Ekbal, "A transformer based multi-task learning approach to multimodal hate speech detection," *Natural Language Processing Journal*, vol. 11, art. 100133, 2025.
- [33] E. F. Ayetiran and Ö. Özgöbek, "An inter-modal attention-based deep learning framework using unified modality for multimodal fake news, hate speech and offensive language detection," *Information Systems*, vol. 123, art. 102378, 2024.
- [34] M. Anand, K. B. Sahay, M. A. Ahmed, D. Sultan, R. R. Chandan, and B. Singh, "Deep learning and natural language processing in computation for offensive language detection in online social networks by feature selection and ensemble classification techniques," *Theoretical Computer Science*, vol. 943, pp. 203–218, 2023.
- [35] P. Yi and A. Zubiaga, "Session-based cyberbullying detection in social media: A survey," *Online Social Networks and Media*, vol. 36, art. 100250, 2023.
- [36] B. Nagy et al., "Privacy-preserving federated learning and its application to natural language processing," *Knowledge-Based Systems*, vol. 268, art. 110475, 2023.
- [37] Y. Khan, D. Sánchez, and J. Domingo-Ferrer, "Federated learning-based natural language processing: A systematic literature review," *Artificial Intelligence Review*, vol. 57, no. 12, art. 320, 2024.

- [38] W. Wang, X. Li, X. Qiu, X. Zhang, V. Brusic, and J. Zhao, "A privacy preserving framework for federated learning in smart health-care systems," *Information Processing & Management*, vol. 60, no. 1, art. 103167, 2023.
- [39] Z. Zhang, Y. Lu, T. Wang, X. Wei, and Z. Wei, "Joint dual feature distillation and gradient progressive pruning for BERT compression," *Neural Networks*, vol. 179, art. 106533, 2024.
- [40] Y. W. Hong, J. S. Leu, M. Faisal, and S. W. Prakosa, "Analysis of model compression using knowledge distillation," *IEEE Access*, vol. 10, pp. 85095–85105, 2022.
- [41] Z. Zhang, L. Wu, D. He, J. Li, S. Cao, and X. Wu, "Communication-efficient and Byzantine-robust federated learning for mobile edge computing networks," *IEEE Network*, vol. 37, no. 4, pp. 112–119, 2023.
- [42] M. G. Herabad, "Communication-efficient semi-synchronous hierarchical federated learning with balanced training in heterogeneous IoT edge environments," *Internet of Things*, vol. 21, art. 100642, 2023.
- [43] Y. Zhou, X. Ma, D. Wu, and X. Li, "Communication-efficient and attack-resistant federated edge learning with dataset distillation," *IEEE Transactions on Cloud Computing*, vol. 11, no. 3, pp. 2517–2528, 2023.